WORKSHEET 24

Introduction to Game Theory

The most basic type of game in (economic) game theory is the *matrix game*. It is played by two players, who will be named Rose and Colin. There is a matrix, and each entry of the matrix contains two numbers. The first number is Rose's payout, and the second number is Colin's payout. Both players simultaneously make their moves: Rose chooses one of the rows of the matrix, and Colin chooses one of the columns of the matrix. Both players wish to get as large of a payout as possible.

For instance, consider the game

Rose \ Colin	L	R
T	$2, 3$	$0, 2$
B	$3, 0$	$1, 1$

Rose can choose the top row (T) or the bottom row (B), whereas Colin can choose the left column (L) or the right column (R). If Rose chooses T and Colin chooses L, then Rose gets a payout of 2, and Colin gets a payout of 3.

PROBLEM 24.1. If both players are playing the above matrix game optimally, what should they do, and what are the payouts?

PROBLEM 24.2. What are the optimal plays in this game?

Rose \ Colin	L	R
T	$3, 2$	$1, 1$
B	$0, 0$	$2, 3$

PROBLEM 24.3. What about this one?

		Colin L	Colin R
Rose	T	$1, -1$	$-1, 1$
Rose	B	$-1, 1$	$1, -1$

The rows (for Rose) and columns (for Colin) are called the *pure strategies* of the game. As you have seen, sometimes there is an optimal selection of pure strategies for each player, and sometimes there isn't. We call a strategy a *Nash equilibrium* if neither player can improve eir payoff, assuming that the other player's strategy remains unchanged.

PROBLEM 24.4. Find all the pure strategy Nash equilibria in the above games.

If the game is going to be repeated multiple times, then the players are not required to play pure strategies: they can use randomness. For instance, Rose could elect to play T $\frac{2}{3}$ of the time and B $\frac{1}{3}$ of the time in some matrix game. We call strategies like this that involve randomness *mixed strategies*. Just like pure strategies, mixed strategies can also be Nash equilibria. The payoff for a mixed strategy is the weighted average of the pure strategies involved in it. For instance, if Rose's payoffs for T and B are t and b, respectively, then in the mixed strategy where she plays T $\frac{2}{3}$ of the time and B $\frac{1}{3}$ of the time, her payoff is $\frac{2}{3}t + \frac{1}{3}b$.

We say that a player is *indifferent* between two (pure or mixed) strategies if they yield the same payoff, when the other player's strategy is fixed.

PROBLEM 24.5. Show that in a mixed strategy Nash equilibrium, both player must be indifferent among all the pure strategies that are played with positive probability.

PROBLEM 24.6. Find all the mixed strategy Nash equilibria in the above games.

PROBLEM 24.7. Suppose that Rose (say) has two pure strategies A and B such that, for any pure strategy by Colin, Rose's payoff in A is strictly greater than that of B. Show that B cannot be a Nash equilibrium. Furthermore, show that if we eliminate B from Rose's options, the Nash equilibria are the same as in the original game.

We call a strategy that is always worse than another strategy a *strictly dominated* strategy. In the matrix, we can cross out all strictly dominated strategies and ignore them. This operation may be performed multiple times. One possibility is that, by using this technique of eliminating dominated strategies, we end up with just a single option for each player. We call this technique iterated elimination of dominated strategies.

PROBLEM 24.8. Can any of the above games be solved by the technique of iterated elimination of dominated strategies?

Colin

		L	R
Rose	T	1 1	0 0
	B	1 1	0 0

PROBLEM 24.9. Construct an example to show that it is not always possible to eliminate *weakly* dominated strategies without eliminating some of the Nash equilibria, where by a weakly dominated strategy we mean the same as a strictly dominated one, but with $\geq$ instead of $>$.

PROBLEM 24.10. Suppose there is a matrix entry where the payoff for each player is higher than that of any other entry. Show that this is a Nash equilibrium. Is it necessarily the only Nash equilibrium?

PROBLEM 24.11. What is the maximum number of Nash equilibria a 2×2 matrix game with all the entries distinct can have? What are all possible such values? Can you classify the circumstances under which each number appears?

PROBLEM 24.12. We can consider ROCK PAPER SCISSORS to be a matrix game, with matrix

Colin

		R	P	S
Rose	R	0, 0	−1, 1	1, −1
	P	1, −1	0, 0	−1, 1
	S	−1, 1	1, −1	0, 0

What are all the Nash equilibria?

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24.1: When Rose thinks about her best choice, she realizes that no matter Colin's choice, choosing Row B will always add 1 to her payout. Colin realizes that Rose will always play B, so he chooses Column R to get a payout of $1>0$. Therefore, both players get 1 point each, despite being able to get more if both deviated from their optimal strategy.

24.2: Here, both player's higher payouts are not lined up with a single choice, so no optimal strategy exists mathematically. However:

- if Rose and Colin could not get 0 payout = (1, 1)
- if both collaborate to alternate rows and columns, then Payout = ((2, 3), (3, 2)).

24.3: No matter both player's selection, either Rose or Colin will have a better outcome for themselves by switching their choice. Therefore, no Nash equilibrium exists in this matrix. In fact, if played many times, the best strategy is to flip a coin so the enemy can't notice any sequence in either Rose or Colin's choice.

24.4: In 24.1, (1, 1) is a Nash equilibrium because either player switching alone leaves them at 0. However, despite (2,3)'s reward, Rose has an incentive to switch to (3,0). In 24.2, despite no optimal play existing, both (3,2) and (2,3) are Nash equilibriums. 24.3 has no equilibriums.

*nash equilibrium

24.5: Assume that a pure strategy is better in payoff then the mixed one. Then, switching strategies would make the player better off, which is impossible given that the mixed strategy is an NE*. Therefore, all pure strategies played with positive ($\neq 0$) probabilities are indifferent to a NE mixed strategy.

24.6: In 24.1, the only optimal strategy leads to a pure NE, so no mixed NEs exist.
In 24.2, the optimal strategy is mixed, so a Nash equilibrium exists mixed. Rose's optimal strategy involves making Colin indifferent between his two pure strategies, which happens with (T=.75, B=.25). Colin's case is symmetrical, so (R=.75, L=.25).

24.6(cont): In 24.3, both players are already indifferent to either strategy, so the NE mixed strategy for both players is (0.5, 0.5), or to flip a coin.

24.7: If B included an NE, then B would somehow have a greater payoff for Rose (to make Rose not switch to A and create an NE), which is impossible. Therefore, all NEs must be with A. Also, because B has no NEs, removing option B for Rose will not change the outcome of the game (Colin will still choose what's best for him anyways, and determine the same NE). In the example of 24.1, Rose always chooses B, and with or without option T, the NE is (1,1).

24.8: Solution for 24.1:

Start by eliminating strategy T, where (2,3) and (0,2) are crossed out, because Rose will always be better off no matter Colin's choice. Then, Colin can choose between a payout of $L \Rightarrow (3,0)$ or $R \Rightarrow (1,1)$. Here, $0<1$, so R is dominant and L is dominated. Therefore, (3,0) and strategy L are crossed out, and (1,1) is the only payout left. This means that (1,1) is an NE. Both 24.2 and 24.3 have no strictly dominant strategies, so no NE exists by process of iterated elimination.

24.9:

		Colin L	Colin R
Rose	T	1,1	0,0
Rose	B	1,1	0,0

Here, R is strictly dominated, but both T and B are weakly dominated despite including both (1,1) NEs. Eliminating T/B removes all NEs!

24.10: If a single value is strictly dominating all other values, then any player changing strategies away from it would make them worse off. That fits the definition of an NE. As for being the only NE, take the famous Stag/Hare example:

Rose \ Colin	Stag	Hare
Stag	2,2	0,1
Hare	1,0	1,1

In this game, both players may want to be safe and choose Hare, but greater rewards are in the stag. Both the 2,2 and 1,1 are NEs, and the mixed strategy NE of (0.5, 0.5) for (Stag, Hare) shows that despite 2,2's dominance, the Stag and Hare are equally balanced benefits.

24.11: Take this example:

Rose \ Colin	L	R
T	2,2	2,2
B	2,2	2,2

Here, 4 nash equilibria exist, and infinite mixed NE's exist, because payoff won't change!

*As in, they are all same (x, y), but $x=y$, $x<y$, $x>y$ is fine.

24.11 (cont.): The above example, with 4 pure NEs, requires all same payouts. Replacing 2,2 with 1,1 BR causes only 3 NEs to exist as both players will move away from 1,1. Again, the three NE payouts must be equal.* In a 2x2 with 2 NEs, either

x, y	SD
x, y	SD

or

x, y	SD
SD	x, y

where NEs x, y weakly dominate the SDs elsewhere on the board, but don't have to be perfectly equal and don't have to dominate diagonally. 1 NE has to weakly dominate every square adjacent but strictly dominate diagonally to avoid a 2nd or 4th NE forming.

24.12: Clearly, no pure strategy Nash equilibria exist here: when a current outcome exists, one player will always have a 0 or -1, and a choice that brings their payout back up to 1. However, a mixed strategy Nash Equilibrium exists: random outcomes, or $1/3$ for each pure strategy. The payoff here is $\frac{1}{3} \cdot 0 + \frac{1}{3} \cdot -1 + \frac{1}{3} \cdot 1 = 0$! However, this is the best any player can do. If Rose bets higher on paper, then Colin will increase his chance to choose Rock. Same for the other 2 strategies, making Rose's random choice truly the best option - and same for Colin. This is the ONLY mixed NE because of this.